

Asmet Ranjan Sahoo

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LinkedIn



GitHub

SUMMARY

Results-driven CS undergraduate focused on Machine Learning, geospatial data engineering, and distributed systems, with hands-on experience building production-ready AI and analytical solutions.

EDUCATION

KIIT University, B.Tech in Computer Science Engineering (3rd Year) Bhubaneswar | 2024 – 2027

B.O.S.E, Cuttack, Diploma in Electronics & Telecommunication Cuttack | 2021 – 2024

Stewart School, Cuttack, Secondary Education (10th) Cuttack | 2009 – 2021

PROFESSIONAL EXPERIENCE

Data Science Intern Apr 2026 – Present

Happiest Minds Technologies | Bhubaneswar

- Built a geospatial MCDM pipeline to identify optimal specialty clinic locations across Delhi and Uttar Pradesh using AHP pairwise comparisons and TOPSIS with Shannon Entropy weighting across 8 spatial criteria.
- Engineered 300m grid cells over OSM-sourced boundaries using OSMnx, GeoPandas, and Shapely; exported ranked site coordinates as CSV for stakeholder reporting.

Tech: Python, OSMnx, GeoPandas, Shapely, Scikit-learn, NumPy, Pandas

Data Science Intern May 2025 – Jun 2025

Terraivise | Bhubaneswar

- Implemented Vision Transformer (ViT + MAE) based semantic segmentation for satellite imagery (Sentinel, Landsat) land-use classification using PyTorch Lightning and GeoPandas.

Tech: Python, PyTorch, GeoPandas, Vision Transformers

Data Science Intern May 2025 – Jun 2025

Pinnacle Labs | Remote

- Developed real-time road lane detection using OpenCV (Canny edge detection + Hough Transform) with robustness across varying lighting conditions.

Tech: Python, OpenCV, NumPy

SKILLS

Languages: Python, C++, SQL (PostgreSQL), Dart, HTML5, CSS3, Javascript

Frameworks/Libraries: FastAPI, Flutter, React, Tailwind CSS, NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn, TensorFlow Lite, OpenCV, GeoPandas, OSMnx, Shapely, LangChain, LangGraph

Cloud & Tools: AWS (EC2, S3), Docker, Git, GitHub, Kafka, Spark, TimescaleDB, Ngrok, REST APIs, Web Scraping

Concepts: OOP, Machine Learning, Deep Learning, Computer Vision, Generative AI, Retrieval-Augmented Generation (RAG), Geospatial Analysis, MCDM (AHP, TOPSIS), ETL Pipelines, Distributed Systems, Data Structures & Algorithms, Agile Methodologies

PROJECTS

KisanDoc – AI Crop Disease Diagnosis App GitHub

- Offline-first Flutter app diagnosing 15 crop diseases across 6 crops; deployed MobileNetV2 (TFLite) achieving ~90% accuracy with <200ms inference on mid-range Android.
- HSV-based severity estimation, multilingual TTS (English, Hindi, Odia), ICAR-sourced treatment plans with pesticide dosage and INR cost estimates.

Tech: Flutter, TensorFlow Lite, MobileNetV2, Hive, Python

Doccy – Medical Emergency Assistance App GitHub

- React Native app for prescription storage and symptom-based emergency guidance with image-based burn severity estimation; backend via FastAPI, deployed on Docker.

Tech: React Native, Python, FastAPI, Docker

Real-Time Cloud Observability & Cost Optimization Platform GitHub

- Distributed cloud monitoring with Kafka + Spark Streaming, anomaly detection on TimescaleDB, containerized and deployed on AWS EC2 with dashboards and alerting.

Tech: Kafka, Apache Spark, TimescaleDB, Docker, AWS EC2

RESEARCH

Soil Moisture Prediction using a Scalable and Validated Random Forest Regressor Model— Regression model using environmental and meteorological features with feature importance analysis for agricultural forecasting applications. [View](#)

CERTIFICATIONS

- IBM Machine Learning Professional Certificate (Coursera, 2025) — Supervised, unsupervised, deep learning and ML capstone. [View](#)
- Health Systems Development: Introduction to Health Systems (Coursera) [View](#)
- Health Systems Development: Health Systems, Policy and Research (Coursera) [View](#)